

Part 12: Assets & Inventory Management Module

Implementation Date:	June 23, 2026
Module:	AssetFeature
Status:	■ Complete
Version:	1.0.0

Executive Summary

The Assets & Inventory Management Module provides comprehensive functionality for tracking and managing corporate assets throughout their lifecycle. This module enables IT administrators to register new assets, allocate them to employees, track their condition, and process returns. Employees can view their allocated assets, request returns, and report damage through an intuitive interface.

Key Features Implemented

Feature	Description	Status
Asset Registration	Register new corporate assets with serial numbers, categories, and conditions	■
Asset Allocation	Allocate available assets to employees with tracking	■
Return Management	Process asset return requests with condition assessment	■
Employee Dashboard	View allocated assets and request returns	■
IT Admin Console	Comprehensive asset inventory management interface	■
Damage Reporting	Flag assets as damaged for IT attention	■
Multi-Category Support	Support for Laptops, Mobiles, Accessories, and Keyboards	■
Status Tracking	Track asset status (Available/Allocated/Returned)	■

Technical Architecture

Backend Components

Domain Layer:

- Asset entity with properties: SerialNumber, AssetName, Category, Condition, Status
- AssetAllocation entity tracking allocation history and returns
- Enums: AssetCategory, AssetCondition, AssetStatus

Application Layer:

- IAssetService interface defining business operations
- AssetService implementing asset lifecycle management

Infrastructure Layer:

- IAssetRepository interface for data access
- AssetRepository with EF Core implementation
- Entity configurations for database mapping

GraphQL Layer:

- AssetFeatureMutation: createAsset, allocateAsset, requestAssetReturn, processAssetReturn
- AssetFeatureQuery: getMyAllocatedAssets, getAllAssets, getAllAllocations

Frontend Components

Employee Asset Dashboard (/assets):

- Grid view of allocated assets with details
- Return request modal with reason selection
- Damage reporting functionality
- Real-time status updates

IT Admin Console (/assets/tracker):

- Tabbed interface (All/Allocated/Available assets)
- Asset registration form
- Quick allocation panel
- Pending returns queue with condition assessment
- Search and filter capabilities

Database Schema

Table	Key Columns	Purpose
Assets	Id, SerialNumber, AssetName, Category, Condition, Status	Store asset inventory
AssetAllocations	Id, AssetId, EmployeeId, AllocatedOn, ReturnedOn, ReturnReason, ConditionOnReturn	Track allocation history

GraphQL API Operations

Operation	Type	Description
createAsset	Mutation	Register a new asset in inventory
allocateAsset	Mutation	Allocate an available asset to an employee
requestAssetReturn	Mutation	Submit a return request for an allocated asset
processAssetReturn	Mutation	Process return and update asset condition
getMyAllocatedAssets	Query	Retrieve employee's currently allocated assets
getAllAssets	Query	Get all assets with optional filters
getAllAllocations	Query	Retrieve complete allocation history

User Workflows

Employee Workflow

- View Assets:** Navigate to /assets to see all allocated assets
- Request Return:** Click "Request Return" and select a reason
- Report Damage:** Use damage report button to flag issues
- Track Status:** Monitor return request status and asset condition

IT Admin Workflow

- Register Assets:** Add new assets with serial numbers and categories
- Allocate Assets:** Search for available assets and assign to employees
- Process Returns:** Review pending returns and assess asset condition
- Monitor Inventory:** Use filters and search to track asset status
- Manage Lifecycle:** Update asset conditions and availability

Implementation Details

Component	Files Created	Lines of Code
Domain Entities	4 files	~80 LOC
Application Services	2 files	~140 LOC
Infrastructure	4 files	~180 LOC
GraphQL Layer	3 files	~110 LOC
Frontend Pages	2 files	~740 LOC
GraphQL Operations	1 file	~110 LOC
Total	16 files	~1,360 LOC

Testing & Validation

Build Verification:

- Frontend build completed successfully with no TypeScript errors
- All components properly typed and validated
- GraphQL operations correctly defined

Integration Points:

- Registered in GraphQLModuleRegistration.cs
- Added to RepositoryRegistration.cs for DI
- Entity configurations applied to ApplicationDbContext

Code Quality:

- Follows established project patterns
- Proper error handling and loading states
- Responsive UI design with Tailwind CSS
- Type-safe GraphQL operations

Future Enhancement Opportunities

- **Asset Maintenance Tracking:** Schedule and track maintenance activities
- **Depreciation Calculation:** Automatic asset value depreciation
- **Barcode/QR Integration:** Quick asset scanning and identification
- **Asset Transfer:** Transfer assets between employees
- **Warranty Management:** Track warranty periods and renewals
- **Asset Analytics:** Utilization reports and cost analysis
- **Bulk Operations:** Import/export asset data in bulk
- **Asset Photos:** Upload and store asset images
- **Notification System:** Alerts for pending returns and maintenance

Conclusion

The Assets & Inventory Management Module has been successfully implemented with comprehensive functionality for tracking corporate assets throughout their lifecycle. The module provides intuitive interfaces for both employees and IT administrators, enabling efficient asset allocation, return processing, and inventory management. The implementation follows best practices with proper separation of concerns, type safety, and responsive design.

All components have been integrated into the existing HRMS architecture and are ready for production deployment.